CPU-First Reinforcement Learning for Native PC-98 Bullet-Hell Games

A Reproducible Systems Study and Research Plan

Touhou PC-98 RL Project

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Abstract

We study reinforcement learning (RL) directly against a native PC-98 bullet-hell game under a CPU-only constraint. The principal systems problem is not merely policy optimization: a useful experiment must boot a legally supplied disk image, identify the correct emulated-memory region, provide isolated resets, extract observations at game rate, and avoid CPU oversubscription. We present an end-to-end TH05 environment that uses exact child-process attachment, compact entity-set observations, a 180,136-parameter recurrent actor–critic, and multi-process proximal policy optimization (PPO). Compact reads have a measured median latency of 0.516 ms, compared with 3.550 ms for the dense-map path, while reducing transferred floats by a factor of 776. Eight emulators deliver about 223 transitions/s on the tested host. After 16,384 training transitions and validation-based checkpoint selection, an untouched four-seed, 900-step paired test improves mean scalar return from 0.0529 to 0.3639 and halves observed deaths. A completion-gated Stage 1 Lunatic curriculum subsequently produces short-phase no-miss clears. Moving the start from phase 3 to phase 2 and fine-tuning for 32,768 transitions improves held-out completion from six to eight of eight and reduces deaths from 19 to 14, but yields no no-miss clear. A later 7,000-step full-stage diagnostic obtains two ordinary clears in four attempts but no no-miss clear. A subsequent control audit finds that one-shot memory input is timing-racy; transactional action refresh and a 1-ms native deathbomb guard cancel three otherwise fatal collisions, after which a fourth collision exhausts the available safety resource. Additional shield-aware PPO and success-only distillation both fail held-out tests. In contrast, a frozen behavior-policy update using successful and failed trajectories improves phase 2–4 held-out no-miss clears from 6/8 to 7/8 and observed deaths from two to zero while changing only the actor head. An action-conditioned future-safety teacher then improves pre-boss held-out no-miss clears from 0/8 to 2/8 and misses from nine to seven; it is removed after distilling the online actor head. On the full stage, the same student increases held-out completion from 4/8 to 7/8, but both policies remain at zero no-miss clears and student misses increase from 12 to 13. The complete one-thread online decision path has 1.474 ms p99 latency, 4.09% of its 36-ms interval. No full-stage no-miss clear has yet been observed. We document negative results, validity threats, and a path toward a common TH01–05 framework.

1 Research questions

This project is organized around seven falsifiable research questions.

RQ1: Systems feasibility. Can an unmodified emulator plus structured memory extraction sustain useful RL throughput on CPU without interfering with other workloads on a shared host?

RQ2: Representation efficiency. Can a compact, permutation-invariant representation of nearby hazards replace large, sparse spatial maps while preserving enough state for bullet-hell control?

RQ3: Learning signal. Under a small CPU budget, does recurrent PPO improve survival and resource management over a paired untrained policy?

RQ4: Safe exploration. Do adapter-certified hard action constraints improve sample efficiency without merely hiding an unsafe policy?

RQ5: Sparse survival credit. Can auxiliary targets or explicit miss costs improve no-miss completion without sacrificing ordinary task completion or injecting a scripted dodge policy?

RQ6: Generality. Which interfaces can be shared by Touhou 1–5, and which memory, termination, reward, and patching components must remain game-specific?

RQ7: Real-time deployment. Can a low-latency online actor and exact safety monitor be separated from a more expensive offline learning plane, and can stronger teachers be distilled without losing survival performance?

The present evidence addresses RQ1 strongly, RQ2 at the systems level, and RQ3 only preliminarily. RQ4 has an underpowered comparative result. An initial RQ5 auxiliary mechanism is rejected by a held-out pilot; a subsequent explicit miss-cost pilot improves the completion point estimate but produces no no-miss clear. RQ6 is deliberately deferred until the agent can complete TH05 on its highest difficulty; only TH05 is currently validated. The transactional actor, native guard, outcome-conditioned offline update, and latency audit provide preliminary evidence for RQ7. Success-only distillation is rejected; a larger future-safety teacher improves the held-out pre-boss point estimate but remains insufficient for a no-miss full-stage clear.

2 Background and related work

The initial reverse-engineering reference describes a high-fidelity, multi-objective bullet-hell environment and an accelerator-oriented baseline [3]. We treat that work as a source for low-level TH05 state discovery, not as the architecture or identity of the present framework.

Our optimizer follows clipped PPO [6] with generalized advantage estimation (GAE) [5]. A game frame contains a variable-size collection of bullets and special projectiles. Processing these objects as sets follows the invariance motivation of Deep Sets [7]; learned attention pooling is related to Set Transformer [2], although our encoder is deliberately smaller and does not perform quadratic self-attention among all hazards.

DOSBox-X supplies PC-98 emulation [1]. ReC98 release P0335 is used only as a clean boot baseline [4]; it is not a runtime or build dependency.

3 Failure analysis of the reference baseline

The reference project solved valuable reverse-engineering problems, but its released implementation encodes assumptions that do not hold for our CPU-only, multi-process, current-emulator setting. We audited those assumptions and reproduced their consequences before replacing them. Table 1 separates observed failure modes from design preferences; "inadequate" here means inadequate for the stated experimental setting, not universally wrong.

3.1 Version-fragile systems assumptions

The released memory walker discards mappings larger than 10 MiB during its initial search and uses an 8 MiB limit for subsequent readable-region filters. On the tested DOSBox-X release, the relevant writable guest mapping is roughly 30 MiB. This is a hard compatibility failure rather than a throughput issue: the correct signature cannot be found in bytes that are never searched. The new 64 MiB cap is still bounded, but its value is checked against the actual mapping layout. Searches are further restricted to writable anonymous regions, retaining a small candidate set.

The original spawn path correctly attaches to its newly created child, but all workers mount one host `export` directory. Its separate attach-existing path chooses the first process returned by a name search. The former risks cross-worker file-state interference and the latter is ambiguous whenever unrelated emulators run on the host. Private writable disk images plus exact child ownership make environment identity explicit. This matters for science as well as reliability: a reset is not reproducible if another worker can alter its persistent state.

The same audit found unbounded coordination waits for workers that remain alive but stop responding, and setup instructions that rely on privileged execution and world-writable build directories. Our runner instead uses child-process tracing, finite deadlines, deterministic cleanup, and ordinary user permissions. These changes reduce both stalled experiments and accidental interference with unrelated jobs.

Memory readability is also weaker than actionability. In a live directional probe, the initial player state was readable immediately but remained fixed for about 80 action intervals while the opening invincibility/input lock was active. The baseline collector records this prefix and attributes its survival

Table 1: Reference-baseline assumptions, observed consequences, and our replacements.

Baseline assumption	Why it fails here	Replacement
XPU/CUDA-only package and dense CNN input	No supported GPU is available; each 2,048-step map rollout was documented as 1.62 GiB	CPU PyTorch lockfile, compact entity sets, and a small GRU
At most 8–10 MiB per scanned memory region	The current DOSBox-X guest mapping is about 30 MiB, so game signatures are never searched	Writable anonymous-region scan with a tested 64 MiB safety cap
Shared host game directory	Concurrent emulators can read and write the same configuration and score files	One private HDI copy and exact child PID per environment
One process-memory input write per action	TH05 refreshes its input word from the physical keyboard every native frame; a one-shot bomb is often overwritten before use	Refresh the selected command every 1 ms during a bounded action transaction
Learner time shares the live game clock	Inference and PPO updates create unobserved game progress and can consume a short deathbomb window	Pause each emulator between transactions; keep the online safety path native and independent of learner latency
Dense $24 \times 92 \times 96$ maps every step	Allocation and language-boundary transfer dominate CPU inference	273-float native compact reader
Normalize every objective before weighting	Reward magnitude loses its semantic meaning and sparse/noisy objectives receive unit variance	Vector GAE, fixed semantic scalarization, then one normalization
Overwrite a latest-only checkpoint	A later update can silently replace a better earlier policy	Periodic snapshots, validation LCB, untouched test
No deadline for a live-but-stalled worker	The learner can wait forever while leaving emulator children behind	Bounded startup/step IPC and exact cleanup
Readable memory implies rollout readiness	About 80 opening action slots ignore control but still emit survival reward	Reset gate waits for the player input lock to end
Patch stage values are human numbered	The loader writes them directly to TH05's zero-based resident stage, silently shifting scenarios by one	Human Stage 1–6 mapped to resident 0–5; both values recorded

reward to sampled actions that the game cannot execute. Our reset gate waits for both an initialized player and the end of this lock. A sustained-left/right probe then moved normalized X from 0.50 to 0 and back beyond 0.62, independently confirming that the subsequent action path is effective.

3.2 CPU cost is determined by representation

The reference package pins an XPU build of PyTorch and states that a GPU is required. More importantly, its learning path stores a $24 \times 92 \times 96$ tensor for every transition, even though most cells are empty. Merely selecting a CPU wheel would therefore leave the principal allocation, copy, memory-bandwidth, and CNN costs intact. Our change is algorithmic as well as infrastructural: nearby hazards are variable-size sets, so a set encoder expresses the symmetry directly and makes CPU inference smaller. Table 2 reports the resulting measured reader cost.

3.3 Optimization semantics and model selection

For objective o, the reference trainer first constructs $\widehat{A}^{(o)} = \mathrm{standardize}(A^{(o)})$, combines these standardized values, and standardizes the sum again. Consequently, multiplying an objective's raw reward by almost any positive constant has no effect, and a rare or noisy objective can be promoted to the same empirical variance as a dense, reliable one. This is not intrinsically invalid, but it no longer implements scalarization in declared reward units. We retain a vector critic and vector GAE, apply explicit semantic weights to the unnormalized advantages, and normalize the resulting policy advantage exactly once.

Finally, overwriting a single "latest" file assumes that training progress is monotone. Our short runs directly contradict that assumption: a policy that improved at 24,576 transitions regressed badly by 49,152. Periodic immutable snapshots, a disjoint validation split, and an untouched paired test do not eliminate selection bias, but they make it visible and prevent the last update from becoming the result by accident. Snapshot ranking is lexicographic: no-miss completion count, completion count, then lower-confidence-bound return. Workers retain episode-local death state so training metrics distinguish a successful clear from a no-miss clear even when an episode spans rollouts.

4 System design

4.1 Reproducible native-game execution

The environment starts from a user-provided Japanese HDI. A fail-closed patch script verifies complete SHA-256 hashes before applying the required executable layouts. The supplied archive is a mixed revision: `MAIN.EXE` requires patch layout 1, whereas `OP.EXE` matches layout 2. Treating layout numbers as a paired version produces a black screen. Patched binaries and game data are never committed.

Each Gymnasium environment copies the 21 MiB template HDI into a private temporary directory. This makes resets independent and prevents concurrent workers from racing on game-written score or configuration files. DOSBox-X is spawned as a direct child, and the native extension attaches to that exact PID. The memory search is restricted to writable anonymous mappings no larger than 64 MiB. This includes the approximately 30 MiB guest-memory mapping used by the tested DOSBox-X release; the previous 8–10 MiB cap silently excluded it.

Startup and step operations have explicit timeouts. Reset retries cover both process creation and the interval before initialized gameplay becomes readable. Workers release inputs and terminate their exact child process on shutdown.

4.2 Compact observation

The compact observation has 273 floats:

$$37 \text{ global features} + 16\times 7 \text{ special-projectile features} + 16\times 7 \text{ bullet features} + 12 \text{ item features.} \tag{1}$$

Entities are ordered by distance for truncation, but the neural encoder pools them as a set. Each entity contains relative position, velocity, type, speed, and normalized distance. Missing entities use the sentinel $(0, 0, 0, 0, 0, 0, 1)$. A previously plausible mask based only on nonzero values is wrong because the sentinel distance is one; the implementation now recognizes the full sentinel and has a regression test for all-padding inputs.

The older observation additionally builds 24 maps of size 92×96, or 211,968 floats per step. Our hot path constructs no map. This is important because map allocation and Python transfer can cost more than policy inference on CPU.

4.3 Policy and value model

Special projectiles and bullets pass through separate shared token networks. Each set is summarized by learned attention pooling and masked max pooling. Global state and item features have small multilayer perceptrons (MLPs); the four summaries are fused into a 128-dimensional vector. A one-layer GRU with 128 hidden units models temporal context. The actor predicts 19 discrete actions. A shared critic head predicts three value components rather than collapsing objectives before value estimation. The plain model contains 180,136 parameters; enabling the two analytic-geometry token inputs used by the present deployment checkpoint raises this to 180,456.

4.4 CPU recurrent PPO

For scaled vector reward $\mathbf{r}_t \in \mathbb{R}^3$, vector GAE is

$$\boldsymbol{\delta}_t = \mathbf{r}_t + \gamma(1 - d_t)\mathbf{V}(s_{t+1}) - \mathbf{V}(s_t), \tag{2}$$

$$\mathbf{A}_t = \boldsymbol{\delta}_t + \gamma\lambda(1 - d_t)\mathbf{A}_{t+1}. \tag{3}$$

The policy advantage is $A_t = \mathbf{w}^\top \mathbf{A}_t$, normalized once after scalarization. This avoids independently normalizing sparse objectives and then assigning them equal empirical variance. PPO uses ratio clipping, value clipping, entropy regularization, gradient clipping, and approximate-KL early stopping.

Rollouts are synchronous across independent emulator processes. Every worker uses one PyTorch thread; the learner defaults to eight. The parent process is CPU-affined and run with positive niceness.

Periodic snapshots are selected on a validation seed set using

$$S = \bar{R} - z\,\frac{s_R}{\sqrt{n}}, \tag{4}$$

after prioritizing observed no-miss stage successes and then ordinary stage successes. Thus the last PPO update is not automatically considered the best model.

Two post-baseline extensions are exposed for controlled ablation. Analytic geometry appends bounded constant-velocity closest-approach features using physical scales supplied by the game adapter. Hard safety masks only actions the adapter can certify as unavailable or behaviorally redundant—currently a bomb with zero stock and movement into an active boundary clamp. The masked categorical is renormalized identically during rollout and PPO optimization, and removed probability mass is logged. It does not implement scripted collision avoidance.

4.5 Real-time control plane and learning plane

The deployment path is intentionally smaller than the research learner. A policy chooses one of 19 actions once per 36-ms transaction. During that transaction, the adapter refreshes the command every 1 ms because TH05's native keyboard scan otherwise overwrites a single memory write. The emulator is paused before returning the next observation, so model inference, replay processing, or a PPO update cannot advance an unobserved game clock. This defines an explicit online control plane even though the present data are still collected on-policy.

An even shorter native safety path reads only `player_is_hit` and the one-byte miss timer. During the audited eight-frame cancellation window it preempts the policy command with bomb. Its response time therefore does not depend on Python inference. The resource-reservation mask runs before geometric filtering; reversing that order allowed a downstream feasibility fallback to spend bombs before a collision. Interventions are exposed in transition metadata rather than silently attributed to the sampled policy.

This boundary permits the offline learner to be more expensive than the deployed actor. Counterfactual replay search, action-contrastive risk, and larger sequence models can label stored states; their movement decisions can then be distilled into the same small recurrent actor. "Offline" here names the learning plane, not a claim that the main PPO implementation is an offline-RL algorithm.

Our first offline control clones attributable actions from complete no-miss episodes and anchors the student to the frozen behavior policy. It tests whether success filtering alone provides useful supervision. The second method retains both positive and negative attempts sampled by a frozen policy. For episode i, utility is

$$u_i = \mathbb{1}[\text{no-miss clear}_i] - n_{\text{miss},i} - 0.05 n_{\text{bomb},i}, \tag{5}$$

and A_i is the centered, standardized utility over the collection batch. If $r_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_b(a_t \mid s_t)$, the actor-head update minimizes

$$-\mathbb{E}_t[\min(r_t A_i, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)A_i)] + \beta\, D_{\text{KL}}(\pi_b \,\|\, \pi_\theta). \tag{6}$$

The encoder, GRU, and critic remain frozen. Native mid-transaction overrides and all bomb actions are excluded from movement-policy credit. This is a small-data conservative policy-improvement experiment, not a claim that an episode-level outcome solves temporal credit assignment.

5 Experimental protocol

Host. The test machine exposes 96 logical CPU threads and approximately 755 GiB RAM. Other experiments occupied selected high-numbered cores. RL jobs were run at niceness 10 and confined to known-idle low-numbered cores. No GPU was visible to PyTorch. The software stack used Python 3.14.6, PyTorch 2.13.0+cpu, and DOSBox-X tag `dosbox-x-v2026.07.02`.

Game condition. The first short-run dataset actually starts at Stage 2 (resident stage index 1) with three lives, three bombs, character 2, rank 0, and power 0. It was initially labelled Stage 1 by following the patch README; source audit showed that the loader assigns `skip_to` directly to TH05's zero-based resident field. We retain and relabel those measurements rather than discard or rewrite them. New scenario tooling accepts human Stage 1–6 and writes resident 0–5. The environment advances at a 36 ms action interval. Evaluations are stochastic and paired by PyTorch sampling seed. The game itself is not yet randomized through a controlled seed API.

Training. The selected policy comes from training seed 202 at update 8: 16,384 total transitions from eight workers. Six snapshots were compared for 900 steps on validation seeds 41, 43, and 44. The final comparison uses untouched seeds 51–54. These test seeds were not used for training or model selection.

Metrics. We report observation latency, transferred floats, model parameters, rollout throughput, scalar return, death count, and the raw resource-reward component. Stage success is the primary task metric but is absent from all short runs. The exact aggregate and per-seed values are tracked in `experiments/2026-08-22-th05-cpu.json`.

6 Results

6.1 Systems efficiency

Table 2: Paused-game observation-reader benchmark.

Reader	Median (ms)	95th percentile (ms)	Floats returned
Dense maps	3.550	5.014	211,968 + 273
Compact	0.516	0.542	273

Compact reading is 6.88 times faster by median and transfers 776 times fewer map-equivalent floats. The compact actor–critic has 180,136 parameters versus 2,375,478 for the comparison CNN–GRU model, a 13.2-fold reduction.

Eight live environments sustain approximately 223 transitions/s when no reset occurs. A 2,048-transition rollout takes about 9.19 s, while three PPO epochs take 0.20–0.33 s. Therefore environment time, not gradient computation, is currently dominant. Increasing learner threads from one to eight improved the model microbenchmark, but 16 threads caused severe oversubscription; bounded threading is part of the algorithmic system, not a cosmetic setting.

6.2 Snapshot selection

Table 3: Validation results for the six candidate snapshots (three seeds, 900 steps each). LCB uses $z = 1$.

Candidate	Mean return	Standard error	LCB score
Reproduction update 8	0.2333	0.2341	-0.0007
Reproduction update 10	0.2112	0.2573	-0.0461
Reproduction update 12	0.0167	0.0145	0.0022
Seed 101 update 8	0.2408	0.2379	0.0030
Seed 202 update 8	0.2743	0.2362	**0.0381**
Seed 303 update 8	0.2769	0.2401	0.0368

Selection first maximizes clears, then no-miss clears, then lower-confidence return. In this experiment all completion counts tie at zero, so seed 202 is selected because it has a slightly higher lower-confidence score than seed 303 despite a lower mean. This illustrates why latest-only or single-seed checkpointing is unreliable at this budget.

6.3 Untouched paired test

The paired mean return gain is 0.3110 with standard error 0.1707. Deaths fall by 50%, and mean resource cost improves by 17.4%. With only four seeds, the uncertainty is large; a normal 95% interval for return gain would cross zero. The correct conclusion is preliminary improvement, not statistical certainty.

6.4 Negative and unstable results

A hand-written short-horizon collision heuristic performs worse than random in the tested condition, reaching four deaths and terminal failure around 948–1,107 steps. Independent nearest-bullet prediction

Table 4: Untouched stochastic test. Resource values closer to zero are better.

Seed	Untrained			Selected PPO		
	Return	Deaths	Resource	Return	Deaths	Resource
51	0.0085	1	-157	0.0135	1	-157
52	0.0165	1	-155	0.6875	0	-87
53	0.1985	1	-83	0.7325	0	-69
54	-0.0120	1	-168	0.0220	1	-152
Mean/total	0.0529	4	-140.75	**0.3639**	**2**	**-116.25**

does not capture multi-projectile escape geometry; the heuristic is retained only as a negative ablation and is not used for behavior cloning.

An earlier exploratory policy improved four-seed return after 24,576 transitions, then regressed badly by 49,152 transitions. Three additional 16,384-transition seeds varied widely. These observations motivated periodic snapshots, validation-based selection, and an untouched test split.

6.5 Matched Lunatic ablation and curriculum probe

Plain compact recurrent PPO and the analytic-geometry plus hard-safety variant were each trained with seeds 1011, 2022, and 3033 for 24,576 transitions. We retained updates 4, 8, and 12, selected nine candidates per condition on seeds 71–73, and tested the selected pair on untouched seeds 81–84 for 1,200 steps. Source audit relabelled this condition as Stage 2 (resident index 1).

Table 5: Matched TH05 Lunatic Stage 2 ablation. LCB is validation mean minus one standard error; deaths are totals across four test seeds.

Condition	Validation mean	LCB	Test mean	Test deaths
Plain PPO	-0.856	-1.449	-1.648	14
Geometry + hard safety	-0.822	-1.422	**-1.054**	**11**

Neither condition clears the stage. The paired test gain is 0.594 with standard error 0.669, and one seed contributes most of it. Thus the point estimate favors geometry plus safety but does not establish a reliable gameplay improvement. The shield constrains 40.1% of test decisions while removing only 2.47% of policy probability on average, consistent with a narrow legality constraint rather than a scripted dodge policy.

A true Stage 1 Lunatic probe at phase 3, ending at phase 4, is cleared by a hard-safety random policy in 397 steps with one death. This confirms that phase termination provides a dense success signal. It motivates a completion-gated curriculum: learn no-miss completion on a short phase, move the starting phase backward, reduce privileged power/resources, and only then optimize a full stage. The machine-readable record is `experiments/2026-08-22-th05-lunatic-ablation.json`.

We then trained two geometry-plus-safety PPO pilots on that phase. Ten retained snapshots were ranked on seeds 111–114. The selected update clears all four runs without a miss in three. Table 6 reports two disjoint eight-seed tests.

Table 6: Stage 1 Lunatic phase 3–4 results. Each row contains eight stochastic episodes; deaths are totals.

Comparison	Clears	No-miss	Deaths	Mean return
Untrained matched policy	8	1	7	1.661
Selected PPO	8	**4**	**4**	**1.952**
Default PPO update 8	8	**3**	**6**	**1.790**
Phase-tuned PPO update 8	8	1	7	1.697

Against the untrained policy, selected PPO gains 0.290 mean return with paired standard error 0.100 and quadruples no-miss clears. This is evidence of learned safety on the short phase, not merely access to an already easy completion. However, the lower-entropy, higher-learning-rate pilot does not replicate its four-seed validation advantage: on seeds 131–138 it trails default PPO by 0.093 return (paired standard error 0.136) and two no-miss clears. We therefore reject any claim that this hyperparameter change is reliably better.

Frequent terminals lower mean collection throughput to 76–80 transitions/s, versus 223 without resets; one rollout with twelve successes takes 64.4 s. This measured 2.8–2.9-fold slowdown strengthens H3: phase curricula need an asynchronous, sequence-aware collector rather than additional learner threads. Exact results are in `experiments/2026-08-22-th05-lunatic-curriculum.json`.

We next froze that checkpoint and evaluated transfer to the longer phase 2–4 task. It clears seven of eight episodes versus six for a matched untrained policy, with 16 versus 19 deaths, but neither condition clears without a miss. The paired return gain is 0.566 with standard error 0.639, so this transfer comparison is inconclusive.

For a clean curriculum transition, weights were loaded while Adam moments, learning rate, update counters, and recurrent state were reset. A 32,768-step default-PPO run produced four snapshots. Validation seeds 301–304 select update 4 by the predeclared clear/no-miss/return-LCB ordering. Although update 16 has higher mean return (2.514 versus 2.373), its higher uncertainty gives a lower one-standard-error LCB (2.333 versus 2.347).

Table 7: Untouched Stage 1 Lunatic phase 2–4 fine-tuning test, eight stochastic episodes per row.

Policy	Clears	No-miss	Deaths	Mean return
Frozen phase 3–4 initializer	6	0	19	1.739
Phase 2–4 fine-tuned PPO	**8**	0	**14**	**2.609**

Table 7 uses untouched seeds 311–318. Fine-tuning gains 0.870 paired return with standard error 0.464. This is evidence for improved completion, not no-miss mastery: every episode still contains at least one death. Curriculum therefore made completion feedback dense but did not solve survival credit assignment. Exact records are in `experiments/2026-08-22-th05-lunatic-p2-curriculum.json` and its linked raw selection and paired reports.

6.6 Sparse-miss auxiliary ablation

We tested the proposed training-only auxiliary target rather than assuming it was beneficial. For each transition, a separate head predicts whether a miss is observed within 15 and 30 actions, conditioned on recurrent state and the sampled current action. Windows stop at episode boundaries; unfinished rollout-boundary negatives are censored, while observed positives remain valid. The balanced binary loss has coefficient 0.05. Future labels and the head are absent at inference. Auxiliary-head initialization restores the PyTorch RNG state so it does not shift policy sampling before the first update.

Plain and auxiliary PPO start from the same selected phase 2–4 checkpoint and each receive 16,384 transitions with training seed 8088. Training totals seem to favor the auxiliary condition (51 versus 102 deaths and four versus thirteen failures), but these trajectories are not controlled because the game RNG is not seeded. Common validation seeds 501–504 select update 4 in each condition. The plain candidate clears four of four with LCB 2.398; the auxiliary candidate clears three with LCB 0.915.

Table 8: Untouched future-miss auxiliary test, seeds 511–518.

Policy	Clears	No-miss	Deaths	Mean return
Plain PPO	**8**	0	**14**	**2.455**
PPO + future-miss auxiliary	7	0	18	2.065

Auxiliary-minus-plain paired return is −0.390 with standard error 0.312. Table 8 therefore rejects this instance of H4: supervision is denser, but completion and deaths both regress and no no-miss clear appears. Vector GAE already propagates a miss reward backward. The auxiliary label is also not counterfactual: it records the outcome under an entire sampled future action sequence without identifying which alternative current action was safer. The lower training death count is thus not evidence of a better frozen policy. Machine-readable results are in `experiments/2026-08-22-th05-lunatic-auxiliary-ablation.json`.

6.7 Explicit miss cost and deployment ablation

The survival reward originally combined a +1 scaled terminal clear, a −0.5 miss, terminal failure, and per-step survival in one critic component. Changing its scalarization weight cannot change the ordering within that component. We therefore detect a miss directly from the resident miss counter and add a configurable miss-only cost after collection. A cost of 2.0 makes a single miss larger than the completion bonus while retaining the old model architecture and weights.

Starting from the selected phase 2–4 checkpoint, one seed receives 32,768 transitions at learning rate 10^{-4} and entropy coefficient 0.005. Validation seeds 621–624 select update 4, which clears four of four but has no no-miss clear. Later updates regress, again validating immutable snapshots.

Table 9: Explicit miss-cost pilot on untouched stochastic seeds 631–638.

Policy	Clears	No-miss	Misses	Mean return
Curriculum initializer	5	0	18	1.403
Extra miss cost, update 4	**7**	0	**16**	**2.099**

The paired return gain is 0.696 ± 0.780 standard error. This favors phase completion but does not solve the target metric: neither policy achieves a no-miss clear. Raw argmax deployment is not a remedy. On separate seeds 641–648, both policies fail 0/8 and incur 24 misses. Their categorical movement distributions are multimodal; repeatedly taking one marginal mode collapses useful stochastic direction switching into a few repeated actions.

We also implemented a policy-accounted emergency-bomb mask: it activates only with remaining bombs, a vulnerable player, and a short predicted close approach. A four-seed calibration produces no no-miss clear and is confounded by uncontrolled game RNG. The approximate geometry omits bullet activation state and replaces TH05's projectile-specific square collision boxes with one Euclidean threshold. We therefore retain the mechanism but do not accept its pilot as evidence. The next shield must follow audited native collision rules and mask unsafe movement actions before falling back to a bomb. Exact pilot values are in `experiments/2026-08-22-th05-lunatic-explicit-miss-cost.json`.

6.8 Audited regular-bullet action mask

We next replace the approximate nearest-object threshold with a native action mask derived from the TH05 collision implementation. Regular bullets use an 8×8 square killbox. A newly grazeable bullet must first enter an asymmetric box extending 16 pixels left, 20 right, and 22 vertically; its killbox becomes active only on a later frame. Delay-cloud and decay states have no hitbox. The adapter combines these states with character-specific focused and unfocused speed and predicts each of the 18 movement actions over six discrete native frames. Bomb remains a policy-accounted action, and special projectiles are deliberately excluded rather than assigned an invented common hitbox.

A four-seed calibration of the frozen miss-cost policy increases clears from 2/4 to 4/4 and reduces total misses from nine to eight, but yields no no-miss clear. We therefore fine-tune under the same mask for 32,768 transitions. Validation seeds 681–684 select update 8 with 4/4 clears, 1.5 mean misses, and mean return 2.727; all four candidate snapshots still have zero no-miss clears.

Table 10: Audited-mask fine-tune on untouched seeds 691–698.

System	Clears	No-miss	Misses	Mean return
Miss-cost initializer	**8**	0	15	2.458
Shield-aware PPO, update 8	**8**	0	**14**	**2.618**

The paired return gain in Table 10 is 0.161 ± 0.150 standard error. This is a small survival improvement and a useful curriculum scaffold, but it is not perfect play: neither system obtains a no-miss clear. The residual deaths motivate auditing deathbomb timing and special-projectile collision rules rather than continuing the same PPO run. Machine-readable results are in `experiments/2026-08-22-th05-lunatic-audited-bullet-safety.json`.

6.9 Real-time action and deathbomb audit

A normal-bomb probe exposes a control failure hidden by the earlier directional test. One write of action 18 leaves the stock at three because TH05's keyboard refresh replaces the command before `player_bomb` observes it. Repeating the same command 142 times over 80 ms consumes exactly one bomb. We therefore replace accumulated wall-clock stepping with a transaction that refreshes the command every 1 ms and freezes the emulator while the learner is inactive.

ReC98 source establishes a 32-frame miss animation and an eight-frame deathbomb interval: a bomb is accepted while the miss timer is 40 through 33, and acceptance clears the pending collision before the resident miss counter increments. The native online guard implements only this audited predicate. In a deterministic-policy diagnostic on the phase 2–4 task, it consumes bombs at steps 194, 509, and 757; all three pending collisions are cancelled without a miss. A fourth collision begins at step 950 with zero bombs and becomes a miss at step 953. Thus the guard converts all available resources into three successful cancellations, but the movement policy still enters four fatal trajectories in 1,200 actions. This is a causal trace under uncontrolled game RNG, not a multi-seed performance estimate. Exact events are recorded in `experiments/2026-08-22-th05-lunatic-realtime-deathbomb.json`.

The result separates two hypotheses. Online latency and action delivery were real failures and are now corrected. Perfect play remains unsolved because a finite emergency resource cannot compensate for repeated unsafe movement. The next policy experiment must reduce pre-collision risk, preferably using a stronger offline teacher or action-contrastive targets distilled into the small online actor.

6.10 Offline movement improvement and online latency

We first test whether simply continuing shield-aware PPO solves the residual movement problem. A 32,768-transition run selects update 4 on validation, but on untouched seeds 731–738 the source obtains six no-miss clears and the update only five; both observe one death. We reject the update. Thus exposing PPO to the corrected online shield does not by itself improve this initializer.

Success-only distillation is also insufficient. Of 24 frozen-policy attempts, 15 are complete no-miss trajectories containing 22,116 attributable movement actions. An actor-only negative-log-likelihood update with a behavior-policy KL anchor reaches 4/4 no-miss on selection seeds, compared with 2/4 for its source. On untouched seeds 851–858, however, both source and student obtain 6/8 no-miss clears and zero deaths; the student's mean return is lower. We reject this method because every sampled action in a successful episode is treated as positive, even when it did not cause success.

The outcome-conditioned update instead retains 32 frozen-policy attempts: 26 no-miss clears and six negative outcomes, including five total deaths. Only the actor head is trained for five offline epochs; epoch 2 is selected on seeds 951–954. Training processes 52,719 attributable actions in 10.15 s, and the selected checkpoint changes no online module. Table 11 shows the disjoint selection and held-out results.

Table 11: Outcome-conditioned actor update. "Clear" includes clears with a miss; checkpoint ranking uses no-miss clear first.

	Selection 951–954			Held-out 961–968		
System	Clear	No-miss	Deaths	Clear	No-miss	Deaths
Frozen source	3/4	2/4	2	**8/8**	6/8	2
Offline actor, epoch 2	**4/4**	**4/4**	**0**	7/8	**7/8**	**0**

The held-out improvement is deliberately stated narrowly: no-miss completion rises by one of eight and two observed deaths disappear, while ordinary completion falls by one because the student has a no-death timeout. This is a better match to the declared perfect-clear objective, not evidence of a large or statistically settled effect. An audit found that the selection tool had previously ranked ordinary completion before no-miss completion; the order and a regression test are now corrected.

Moving the boundary earlier changes the conclusion. On the Stage 1 phase 1–4 task, the frozen source obtains 8/8 held-out no-miss clears whereas the same outcome update obtains only 5/8, one miss, and two no-death timeouts. We reject that update. A 7,000-step diagnostic from the actual beginning of Stage 1 then obtains two ordinary clears in four attempts but no no-miss clear and six misses. Five of those six misses occur before a boss phase is active. Isolating the pre-boss phase 0–1 segment confirms the boundary: the source completes all eight attempts but only two without a miss.

We therefore train a more local offline teacher without adding it to the online path. From 24 frozen-policy pre-boss trajectories, let e_t denote an exact native collision preemption or an observed miss. For horizons $h \in \{8, 24, 64\}$, the training-only action-conditioned risk head predicts

$$y_{t,h} = \mathbb{1}[\exists k \in \{1, \ldots, h\} : e_{t+k} = 1] \,. \tag{7}$$

The current preempted action is excluded, unfinished suffixes are censored, and terminal safe suffixes remain valid negatives. Balanced binary cross-entropy processes 290,988 valid labels, including 10,522 positives. At epoch 5, mean predicted probability is 0.513 on positive labels and 0.442 on negative labels. The teacher evaluates all 19 actions offline and forms a risk-adjusted distribution from behavior logits minus three times predicted 64-step risk. Only the actor head is distilled toward that distribution, with a behavior-policy KL anchor of 0.5; the selected epoch changes behavior KL by approximately 3×10^{-4}.

Table 12: Pre-boss future-safety distillation. Misses are totals and every episode completes the phase 0–1 segment.

	Selection 1451–1454			Held-out 1461–1468		
System	Clear	No-miss	Misses	Clear	No-miss	Misses
Frozen source	4/4	2/4	2	8/8	0/8	9
Distilled actor, epoch 4	4/4	**3/4**	**1**	8/8	**2/8**	**7**

Table 12 is a directionally positive held-out result, not a causal action-risk estimate. The teacher observes only behavior actions and then extrapolates to alternatives; the small probability separation and sample count make counterfactual emulator branching the appropriate next test. A negative hard-safety control also matters: replacing the audited H6 mask's empty-intersection fail-open behavior with a latest-predicted-collision fallback produces 0/4 no-miss clears versus 2/4 for the unchanged mask. H10 and H16 variants tie the baseline's 2/4 no-miss count but constrain more actions and have lower return lower-confidence bounds. We reject all three fallback variants.

The deployment exporter retains only recurrent model weights, runtime arguments, and provenance. It removes the optimizer, the future-safety head, and trajectory paths, reducing the checkpoint from 939,122 to about 742,600 bytes. Thus the richer learner changes the online actor weights but not its architecture or decision-time computation.

We also implement the substrate for the next, counterfactual teacher. Each offline emulator receives a private DOSBox-X save file and is controlled through the X11 window belonging to its exact PID under Xvfb. MAIN.EXE's live `_stage_frame` counter is used to stop after an exact number of guest frames; the resident run-statistic previously exposed under that name remained zero during play and is corrected. In an early pre-boss probe, restoring an anchor reproduces the 273-float observation and serialized raw game state exactly. Repeating one action for eight two-frame decisions reproduces every intermediate frame number and both final SHA-256 digests, with zero maximum observation error. A contrasting horizontal action changes raw state and the observation by up to 0.375. This validates deterministic branching, not a survival improvement; high-risk collision labels and distillation remain to be tested.

The first high-risk probe shows why trigger semantics matter. Table 13 holds the first action for the real two-native-frame decision interval and then uses the same deterministic actor continuation without bombs. H6 marks one action unsafe under a six-frame constant-velocity projection, yet none of the 12 structurally legal actions collide over the 26-frame branch. We reject H6 exclusion as a current-action label. At an H2 trigger, by contrast, ten of 12 legal actions enter the exact pending-hit window within two or four native frames, while actions 3 and 12 survive the complete continuation. The emulator state, recurrent hidden state, continuation controller, and timing are matched; only the first action differs.

Table 13: Single-anchor offline counterfactual feasibility probes.

Trigger	Stage frame	Legal	Collision	Safe	Earliest hit
H6 held-motion risk	427	12	0	12	–
H2 immediate risk	439	12	10	2	2 frames

This is causal-label feasibility, not policy improvement. A multi-trajectory, group-held-out branch dataset and a distilled-actor evaluation are required before the labels count as learning evidence.

We next tighten the deployment objective to no miss, no bomb (NMNB). This is a different task from the earlier deathbomb-enabled evaluations: action 18 is permanently masked, the native deathbomb guard is disabled, and a success requires a stage clear, zero misses, and zero bomb actions. Five trajectories collected before this clarification are excluded because their navigation histories could contain automatic deathbombs. This exclusion illustrates why the action-space contract belongs in checkpoint and dataset provenance rather than in an informal evaluation description.

For the clean dataset, seeded categorical source-policy navigation diversifies states between anchors, while every branch continuation remains deterministic H6 control without bombs. We retain only complete anchors containing at least one colliding and one horizon-safe movement action. Complete trajectories are assigned to train, selection, or held-out groups; no compact-observation hash occurs in two groups. Table 14 summarizes the resulting 39 anchors.

Table 14: Grouped H2-triggered, H6-continuation NMNB counterfactual data.

Split	Trajectories	Anchors	Collision actions	Safe actions
Train	6	18	52	248
Selection	3	9	27	117
Held-out	4	12	52	152

The direct teacher subtracts three times binary collision risk from behavior logits and retains a 0.5 behavior KL anchor. Only the four actor-head tensors are trainable; encoder, GRU, and critic tensors remain byte-identical. The best nonzero epoch lowers train expected collision mass from 0.19159 to 0.19108, but raises selection mass from 0.16904 to 0.16928 and held-out mass from 0.23012 to 0.23028. The corresponding risk-adjusted objectives also worsen on both untrained groups. Crucially, epoch zero participates in selection, so the scientific decision is to retain the source. Epoch one is kept only as a minimum-change diagnostic student; forcing a trained epoch to win would turn a negative experiment into selection bias.

Table 15 returns to the complete Stage 1 image under the same NMNB action mask. Neither system clears any episode and every episode consumes all three lives. The diagnostic student has a higher held-out mean return and median survival, but slightly lower mean survival and identical failure count. Held-out results do not override the earlier source selection.

Table 15: Complete Stage 1 NMNB evaluation; misses are totals and bomb actions are zero in every episode.

	Selection 1861–1864				Held-out 1871–1878			
System	Clear	NMNB	Miss	Mean steps	Clear	NMNB	Miss	Mean steps
Frozen source	0/4	0/4	12	1775.5	0/8	0/8	24	2238.8
Diagnostic actor	0/4	0/4	12	2216.2	0/8	0/8	24	2216.6

Thus exact action credit is operational, but the present 18-anchor update does not generalize through a frozen recurrent representation. The next test must increase stage-stratified coverage and measure action-risk separability before unfreezing representation layers. The online/offline boundary remains intact: the diagnostic actor adds no module, and its one-thread complete-path p99 is 1.535 ms, or 4.26% of the 36-ms budget.

The full-stage result separates completion from survival. Table 16 uses a 7,000-step limit and the same explicit runtime safety configuration for both checkpoints. An earlier attempted comparison inherited different deathbomb settings from checkpoint metadata; it was terminated and excluded, and the selector now rejects such mismatches unless the comparison is marked as an intentional runtime ablation.

Table 16: Complete Stage 1 Lunatic future-safety evaluation. Misses are totals; no condition obtains a no-miss clear.

	Selection 1491–1494			Held-out 1501–1508		
System	Clear	No-miss	Misses	Clear	No-miss	Misses
Frozen source	2/4	0/4	7	4/8	0/8	**12**
Distilled actor	2/4	0/4	**5**	**7/8**	0/8	13

The 4/8 to 7/8 held-out completion gain is useful for moving a curriculum boundary, but it is not progress on the declared perfect-clear metric: both systems remain at 0/8 no-miss and the student's total misses increase by one. Longer completed trajectories also change exposure to late patterns, so raw miss totals and completion must both remain visible. We retain this actor as a completion initializer and reject the observational teacher as a sufficient survival-credit solution.

Finally, we benchmark the selected student's complete decision path on one CPU thread using a paused live-game state, 500 warm-up decisions, and 5,000 measured decisions. Table 17 includes the audited masks, recurrent forward pass, and categorical sampling. The 1.474-ms complete-path p99 consumes 4.09% of the 36-ms interval. In the transactional harness, inference occurs while the emulator is paused, so even a tail outlier cannot advance hidden game time; the benchmark additionally shows sufficient headroom for an unpaused deployment.

Table 17: One-thread online latency in milliseconds over 5,000 decisions.

Component	Mean	p50	p95	p99	Max
Native/audited shields	0.035	0.034	0.038	0.050	0.073
Policy and sampling	1.344	1.341	1.385	1.440	2.787
Complete decision	1.379	1.376	1.421	1.474	2.830

Exact datasets, seeds, checksums, negative controls, and latency quantiles are recorded in five machine-readable files:

- `experiments/2026-08-22-th05-lunatic-offline-online.json`;
- `experiments/2026-08-22-th05-lunatic-stage-boundary.json`;
- `experiments/2026-08-22-th05-savestate-branching-probe.json`;
- `experiments/2026-08-22-th05-high-risk-counterfactual-pilot.json`;
- `experiments/2026-08-22-th05-immediate-counterfactual-pilot.json`;
- `experiments/2026-08-23-th05-nmnb-counterfactual-distillation.json`.

7 Threats to validity

- The full-stage selection comparison has four stochastic seeds and its held-out comparison has eight. Standard errors remain material, and no full-stage no-miss clear occurs.
- Evaluation seeds control action sampling, but the native game does not yet expose a separately controlled environment RNG seed.
- The outcome-conditioned update uses only 32 behavior-policy episodes and assigns one utility to every attributable movement in an episode. Its 7/8 versus 6/8 held-out no-miss result is a promising pilot, not resolved long-horizon causal credit or a confidence-bounded improvement claim.
- The future-safety head receives 24 behavior-policy trajectories and only factual chosen-action outcomes. Evaluating its score on all 19 actions is observational extrapolation, not counterfactual supervision; 2/8 versus 0/8 pre-boss held-out no-miss clears remain an underpowered point estimate.
- Exact replay is verified at an early pre-boss anchor and action-dependent outcomes at one immediate-risk anchor. Two trigger probes do not establish multi-state coverage or improvement in a distilled actor.
- The comparison model has no reproducible compatible checkpoint, so systems comparisons do not imply a gameplay win over prior work.
- Learning evidence now includes the start of TH05 Stage 1 at low power and complete Stage 1 evaluation, but training remains segmented and uses one character and rank. It does not establish full-stage no-miss mastery, later stages, other characters or ranks, or TH01–04.
- The compact observation drops most hazards beyond the nearest 16 in each category and currently omits a compact normal-enemy set.

- Host timing is measured while unrelated jobs are present. CPU affinity and niceness reduce interference but do not produce a dedicated-machine benchmark.
- API steps are transactional and the emulator is paused during learner work, but each live interval is still paced by wall clock rather than an observed native-frame counter. Setting the interval to zero is rejected and cannot be interpreted as additional environment transitions.

8 Research roadmap

The next experiments are specified as hypotheses rather than assumed upgrades.

H1: richer compact geometry. Evaluate the implemented closest-approach features, then add normal-enemy tokens while keeping the observation below 2 KiB. Compare each addition against the 273-float schema with matched training transitions and validation/test splits.

H2: TH05 Lunatic mastery. Train a stage/phase curriculum that ends at rank 3 with no privileged starting resources. Promote a policy only after stage completion, then optimize no-miss completion and finally full-game completion. Use completion and resource-normalized completion as primary metrics rather than short-prefix return.

H3: reset-aware collection. When terminals become common, replace global synchronous waiting with an asynchronous collector or reset pool. The prediction is improved wall-clock throughput without changing the on-policy age bound.

H4: sparse-miss credit (two incomplete pilots). The tested on-policy 15/30-step prediction loss reduces neither held-out deaths nor completion failures. Do not tune it on the reported test seeds. A new pre-registered experiment must add actionable information—for example, action-contrastive short-horizon risk or an explicit survival constraint—and use multiple training seeds at matched transitions. It is rejected if no-miss completion does not improve or ordinary completion regresses.

An explicit miss-only cost improves the completion point estimate from 5/8 to 7/8 but yields zero no-miss clears in both conditions. It remains available as an objective option, but is not sufficient evidence for H4.

H5: risk-aware objectives. Compare the fixed scalarization with constrained or distributional survival objectives. Any method must report the full reward vector and may not hide a survival regression behind score gain.

H6: adapter-certified safety. Compare no mask, exact resource/boundary constraints, and any future predictive shield at matched training budgets. Predictive masks must encode audited bullet activation and projectile-specific collision boxes, not a single nearest-object distance. Report removed policy probability mass and evaluate every trained policy both with and without its shield.

The audited regular-bullet mask improves a four-seed completion calibration and reduces untouched-test misses from 15 to 14, but both systems remain at 0/8 no-miss. H6 is therefore partially supported as a curriculum mechanism and rejected as a sufficient perfect-play solution. The next registered extension must attribute deaths and add only audited special-projectile or deathbomb semantics.

H7: offline teacher and online actor. Keep the deployed recurrent actor and native safety monitor within the measured control deadline. Permit replay search or a larger sequence teacher only in the paused/offline learning plane, then distill action distributions and action-contrastive risk into the online actor. Compare the distilled actor against its source policy on held-out no-miss completion, intervention count, and inference latency; reject any teacher improvement that disappears after distillation.

The success-only student is rejected because its 4/4 selection result falls to the source's 6/8 no-miss rate on held-out seeds. A positive-and-negative outcome update provisionally passes: it moves held-out no-miss clears from 6/8 to 7/8 and removes two observed deaths without changing the online architecture. H7 is therefore supported as an architectural separation and remains open as a learning hypothesis. At the earlier pre-boss boundary, the observational future-safety teacher moves held-out no-miss clears from 0/8 to 2/8 and misses from nine to seven. On the complete stage it increases held-out clears from 4/8 to 7/8, but both systems obtain zero no-miss clears and student misses rise from 12 to 13. Its exported online path remains small at 1.474-ms p99. The deterministic save-state substrate passes byte-exact replay, and the H2 pilot yields ten colliding versus two safe legal actions from the same high-risk anchor. The grouped NMNB follow-up collects 39 anchors from 13 disjoint trajectories. Every nonzero actor epoch loses to the source on selection objective, and the diagnostic epoch obtains 0/8 held-out Stage 1 clears with 24 misses, identical to the source. H7 remains supported as a systems boundary but this first counterfactual learner is rejected. The branch evaluator and save states remain absent from deployment; the next test must scale stage-stratified data and audit frozen-feature action-risk separability.

H8: old-five-games interface. Define per-game contracts for observation schema, action meaning, terminal flags, executable identity, and reset. TH01–04 enter the benchmark only after live parser/action tests pass and TH05 mastery has been addressed; shared code alone is not evidence of support.

9 Conclusion

Native PC-98 RL is practical on a CPU when observation construction, process attachment, worker isolation, and thread scheduling are treated as first-class research concerns. The present system removes a large dense-map bottleneck, runs eight independent games at useful speed, and exhibits small-data improvements in short-phase completion and survival. A future-miss auxiliary loss fails a held-out pilot, and an explicit miss cost improves ordinary completion without producing a no-miss clear. An audited regular-bullet mask then preserves 8/8 completion and reduces misses from 15 to 14, but again produces zero no-miss clears. Transactional input and a 1-ms online deathbomb guard then cancel three consecutive fatal collisions, but a fourth occurs after resources are exhausted. Shield-aware PPO and success-only distillation fail their held-out tests. A conservative offline update using positive and negative episode outcomes then improves phase 2–4 held-out no-miss clears from 6/8 to 7/8 and deaths from two to zero while leaving the online architecture unchanged. At the true Stage 1 start, a training-only future-safety teacher distilled into the actor improves held-out pre-boss no-miss clears from 0/8 to 2/8 and complete-stage clears from 4/8 to 7/8. It does not solve survival: both full-stage policies have zero no-miss clears and student misses increase from 12 to 13. The exported 180,456-parameter actor contains neither optimizer nor risk head and its complete decision p99 is 1.474 ms on one CPU thread. This is evidence for the offline/online boundary and curriculum completion, not mastery. Byte-exact save-state replay then supports a strict NMNB dataset of 39 anchors from 13 trajectory-disjoint runs. The causal labels are real, but an 18-anchor actor-head training split does not generalize: epoch zero wins selection, and both source and diagnostic student obtain 0/8 held-out Stage 1 clears with 24 misses. The main scientific task is now stage-stratified counterfactual coverage and a frozen-feature separability audit, followed by the omitted normal-enemy observation and eventually TH01–04 adapters.

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